

Pre-1910 Well Measurements in the United States

Metadata

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S1. Early 1900s data compilation and quality control

We compiled thousands of water level measurements reported in tables within US Geological Survey reports published in the early 1900s: (a) Crider & Johnson, 1906 (b) Darton, 1905 (c) Fuller et al., 1905 (d) Fuller & Sanford, 1906 (e) Veatch, 1906*.

Our search for data sources took place from May-June 2021; our data compilation and transcription took place from June-October 2021; our data cleaning occurred in November 2021.

*our transcription of Veatch, 1906 is incomplete in the state of Texas. The following counties were not transcribed: Anderson, Angelina, Camp, Cherokee, Franklin, Gregg, Hopkins, Houston, Tasper, Morris, Polk, Red River, Rusa, Sabine, San Augustine, Shelby, Smith, Titus. See Search Criteria below for details. All other sources were transcribed in their entirety.

1.1 Search criteria

We searched for publications from the U.S. Geological Survey that were available online in the U.S. Geological Survey Publication Warehouse. We specifically looked for water level measurements that were made prior to 1910 and recorded in a tabular format.

Our pre-1910 data compilation is not comprehensive. There are other early well water level measurements available in different U.S. Geological Survey publications which we did not hand digitize due to limited time and resources. For the purpose of our study, we sought early records that spanned the continental US (i.e., Darton, 1905, which contains measurements from all 48 continental US states), as well as some documents that focused on regional aquifer systems (i.e., Crider and Johnson, 1906, which focuses on the Mississippi Embayment Regional Aquifer System).

1.2 Data sources

- (a) Crider, A. F. & Johnson, L. C. *Summary of the underground-water resources of Mississippi. Water Supply Paper* <https://pubs.er.usgs.gov/publication/wsp159> (1906) doi:[10.3133/wsp159](https://doi.org/10.3133/wsp159).
- (b) Darton, N. H. *Preliminary list of deep borings in the United States. Preliminary list of deep borings in the United States* vol. 149 180 <http://pubs.er.usgs.gov/publication/wsp149> (1905).

- (c) Fuller, M. L., Lines, E. F. & Veatch, A. C. *Record of deep-well drilling for 1904. Record of deep-well drilling for 1904* vol. 264 106 <http://pubs.er.usgs.gov/publication/b264> (1905).
- (d) Fuller, M. L. & Sanford, S. *Record of deep-well drilling for 1905. Record of deep-well drilling for 1905* vol. 298 299 <http://pubs.er.usgs.gov/publication/b298> (1906).
- (e) Veatch, A. C. *Geology and underground water resources of northern Louisiana and southern Arkansas. Geology and underground water resources of northern Louisiana and southern Arkansas* vol. 46 480 <http://pubs.er.usgs.gov/publication/pp46> (1906).

1.3 Transcription quality check

Five students at the University of California, Santa Barbara helped to manually transcribe the early 1900s well data (see Acknowledgements in the main text). Students were instructed to follow similar data compilation protocols. The data were transcribed, checked twice by the transcriber, then subjected to a final review for their accuracy by lead-author A.E. Hilton.

1.4 Data cleaning

The following modifications were made to the original text during transcription:

- (i) punctuation marks in text (e.g., "?", "!", "") were removed
- (ii) fractions were replaced with their decimal equivalent (e.g., $\frac{1}{2}$ becomes 0.5)
- (iii) blank columns indicated by punctuation (e.g., ".....") were replaced with NA
- (iv) "do" in text indicated "same as above" and was replaced by text preceding it
- (v) columns were renamed for brevity and to be consistent with other data sources (see Columns below)

1.5 Columns

Table S1. Original and modified columns of early 1900s data.

Data Source	Original Column	Modified Column *
Crider & Johnson, 1906	No	id
	State	state
	County	county
	Town	town
	Owner	owner
	Township	township
	Range	range
	Section	section
	Year completed	year
	Diameter of well	diameter_in
	Depth of well	min_depth_ft
		max_depth_ft
		depth_sign
		depth_text
	Depth to principal water supply	depth_water_supply
	Height of water above (+) or below (-) mouth of well	min_height_water_ft
		max_height_water_ft
	How obtained at surface	obtained

Data Source	Original Column	Modified Column *
	Quality	quality
	Supply per minute	yield_per_min_gal_pump
	Increase or decrease of supply	increase_decrease_supply
	Effect of pumping on level of water	pumping_effect
	Geologic horizon of well mouth	geologic_well_mouth
	Geologic horizon of principal water-bearing stratum	geologic_water
		citation
		page_no
Darton, 1905		id
	Location	state
		county
		city
		specific_location
	Depth	min_depth_ft
		max_depth_ft
		depth_sign
	Diameter	diameter_in
	Yield per minute	yield_per_minute_gallons
	Height of water	min_height_water_ft

Data Source	Original Column	Modified Column *
		max_height_water_ft
		height_water_text
	Remarks	remarks
		citation
		page_no
Fuller et al., 1905	No	id
	Location	state
	County	county
		city
		specific_location
	T	township
	R	range
	S	section
	Owner	owner
	Contractor	contractor
	Driller	driller
	Authority	authority
	Depth	min_depth_ft
		max_depth_ft

Data Source	Original Column	Modified Column *
		depth_sign
		depth_text
	Diameter	diameter_in
	Depth to principal water or oil supply	depth_water_supply
	Height of water	min_height_water_ft
		max_height_water_ft
		height_water_text
	Yield per minute Flow	yield_per_min_gal_flow
	Yield per minute Pump	yield_per_min_gal_pump
	Year completed	year
	Kind of well	kind_of_well
	Remarks	remarks
		citation
		page_no
Fuller & Sanford, 1906	No	id
	Kind of well	well_type
	Location	state
	County	county
		city

Data Source	Original Column	Modified Column *
		specific_location
	Owner	owner
	Contractor	contractor
	Driller	driller
	Authority	authority
	Depth	min_depth_ft
		max_depth_ft
		depth_sign
		depth_text
	Diameter	diameter_in
	Depth to principal water or oil supply	depth_water_supply
	Height of water	min_height_water_ft
		max_height_water_ft
		height_water_text
	Yield per minute Flow	yield_per_min_gal_flow
	Yield per minute Pump	yield_per_min_gal_pump
	Year completed	year
	Remarks	remarks
		citation

Data Source	Original Column	Modified Column *
		page_no
Veatch, 1906	No	id
	Location	state
		county
		city
		specific_location
		township
		range
		section
	Owner	owner
	Driller	driller
	Authority	authority
	Diameter of well	diameter_in
	Depth of well	min_depth_ft
		max_depth_ft
		depth_sign
		depth_text
	Approximate elevation of surface	elevation_ft
	Height of water above (+) or below	min_height_water_ft

Data Source	Original Column	Modified Column *
	(-) the ground	max_height_water_ft
		height_water_text
	Depths of principal water bearing strata	min_depth_ft_strata
		max_depth_ft_strata
	Yield per minute Flow	yield_per_min_gal_flow
	Yield per minute Pump	yield_per_min_gal_pump
	Geologic horizon of water bearing strata	geologic
	Quality	quality
	Remarks	remarks
		citation
		page_no

*All headings are shorthand for the original column headings; some headings were added for clarity. Multiple headings for location were added if there was more than one locator description (e.g., separating “city”, “state”, “township”, “range”, “section”, and any shorthand location descriptions as “specific_location”). Multiple headings were also created to accurately track original data entries and include units of measurement, which often included numerical ranges, text, and special characters. These modifications include: (i) “Depth of well” which becomes “min_depth_ft”, “max_depth_ft”, “depth_sign”, “depth_text”; (ii) “Height of water above (+) or below (-) the ground” which becomes “min_height_water_ft”, “max_height_water_ft”, “height_water_text”; (iii) “Depths of principal water bearing strata” becomes “min_depth_ft_strata”, “max_depth_ft_strata”; (iv) “citation” and (v) “page_no” headings were added to denote the citation of data source and the page number of the data reference.

Table S2. Columns of combined early 1900s data.

Complete early 1900s records	id
	state
	county
	city

	specific_location
	township
	range
	section
	latitude*
	longitude*
	min_depth_ft
	max_depth_ft
	depth_sign
	depth_text
	min_height_water_ft
	max_height_water_ft
	height_water_text
	remarks
	citation

*columns added with geolocation

1.6 Geolocation

No wells in the early 1900s US Geological Survey reports have an associated latitude and longitude coordinate recorded within the report; the location of these wells are often provided as a description (e.g. State, County, City) or a township, range, and section. We estimated the latitude and longitude coordinates of each unique well by the following steps:

- (1) Remove wells with no location information.
- (2) Separate wells with only township, range, section (no city, county, state information)
- (3) Use “Geocode by Awesome Table” software to locate based on city, state column for remaining wells.
- (4) Check all geolocations created in step 3 for accuracy, correct NA values or locations unable to be found.
- (5) For wells with only a township, range, and section (step 2), locate township, range, section and calculate latitude and longitude based on the centroid of that township, range, section.
- (6) For wells with a "specific location" (additional location information beyond township, range, section, city, county, state), estimate specific location if possible and add latitude/longitude estimation. For example, a “specific location” (as defined here) might read as follows: “6m nw of”. We used the geocoding software to estimate the landmark or city of the well (e.g., "Jackson, Alabama") and then applied the horizontal offset (e.g., 6 miles north west of Jackson, Alabama) to estimate the latitude and longitude of the well.

1.7 Artesian indications

In the pre-1910 dataset, the depth to water measurements for each well are not described consistently. For example, some depth to water measurements are recorded as "below the mouth of the well," "below land surface", or simply as "height of water". We highlight that different early 1900s reports have different measurement reference points and/or measurement practices.

Table S3. Artesian conditions were indicated either by a positive number (+) or written text.

Sign	+
Text	"flow", "flows", "one flows", "flowed at first", "flowed", "flowed once", "slight flow", "flowed originally", "to surface", "nearly to surface", "rises to surface", "near surface", "surface", "at top", "nearly to top"

1.8 Data used in analyses

Data used in this study primarily included well location (estimated latitude and longitude), depth of well, height of water, and an artesian indication (by text or sign). Where ranges existed for well depth and height of water (see Columns above, e.g., indicated by "min_depth_ft", "max_depth_ft"), we utilized the maximum value.

We excluded well water level measurements from further analyses if (a) the well was identified as an oil and gas well, (b) the record did not specify a numeric well depth, (c) there was no recorded water level measurement or water depth indication, or (d) the recorded well location could not be straightforwardly converted into an estimate of the latitude and longitude, such as short-hand locations (e.g., Spence Run, Allegheny County, Pennsylvania).

1.9 Notes of limitations

Sometimes wells recorded in these historical documents are not one specific well, but are meant to represent multiple wells. Among the reports we consulted, this notation (i.e., multiple wells captured by one row in a table) is most common in Darton, 1905, pg. 7 where it is stated "In regions of oil and gas wells, where borings are numerous, the individual wells cannot be listed here, but representative wells are given."